

1) Why python is dynamically typed language

Python is called a **dynamically typed language** because you do not need to declare the data type of a variable when creating it. Python decides the type **automatically at runtime** based on the value assigned to the variable.

For example:

```
x = 10
```

Here, Python understands that x is an integer.

Later, you can assign a string to the same variable:

```
x = "Hello"
```

Now x is a string. You don't need to specify int or str.

So, in simple words:

Python is dynamically typed because the variable's data type is determined at runtime and the same variable can hold values of different data types at different times.